



THIRUKUMARAN S

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CAREER OBJECTIVE

To start my career as a Forward Deployed Engineer in a reputed organization where I can leverage my technical knowledge, learn from real-world challenges, and contribute to building innovative, customer-focused solutions while growing professionally.

EDUCATIONAL QUALIFICATION

Dr. N.G.P Institute of Technology, B.Tech **2023–2027**

(CSBS) CGPA: 7.77 (up to 5th semester)

APS Academy Matric Higher Secondary School, **2022–2023**

HSC Score: 90.33%

PROJECTS

AI-BASED HUMAN ACTIVITY RECOGNITION SYSTEM **APR 2026**

- Developed a real-time Human Activity Recognition (HAR) system using CCTV video feeds, OpenCV, and pose estimation models to monitor human activities in security-sensitive environments.
- Implemented rule-based activity classification to detect suspicious behaviors such as loitering, unauthorized access, and abnormal movements, with automated alert generation.
- Designed a CPU-efficient, cost-effective, and locally deployable solution that operates without cloud dependency, enhancing privacy and real-time security monitoring.

TECH STACK : Python, OpenCV, YOLOV11 Pose, NumPy, CCTV Camera Integration.

AI-BASED PEDESTRIAN ROAD CROSSING SYSTEM **SEP 2025**

- Developed an AI and IoT-based smart traffic management system to enhance pedestrian safety by detecting pedestrians using OpenCV, CCTV cameras, and sensors.
- Integrated real-time environmental sensing and pedestrian detection to monitor road conditions, vehicle movement, and crossing requests for safer traffic control.
- Implemented adaptive smart traffic lights that dynamically adjust signal timings based on AI-driven analysis, improving traffic flow and reducing pedestrian risk.

TECH STACK : Python, OpenCV, Arduino Uno, YOLOV8, Traffic Light Module, NumPy, Arduino IDE.

THE QUIZ **DEC 2024**

- Built a web-based quiz management system for creating, managing, conducting online assessments.
- Enabled user authentication, support for multiple question types, and automated grading.
- Provided result analysis and performance tracking features to deliver instant feedback and assessment insights.

TECH STACK: HTML, CSS, JavaScript, MySQL, Bootstrap.

INTERSHIP

Machine Learning – Nitroware Technologies

2025

- Developed and trained Machine Learning models using Python, NumPy, and Pandas for data-driven problem solving.
- Performed data preprocessing, feature engineering, and model evaluation using supervised and unsupervised learning techniques.
- Analyzed model performance and generated insights through experiments conducted in Google Colab.

TECH STACK: Python | Google Colab | NumPy | Pandas

AREA OF INTEREST

- Prompt Engineering
- Software Development
- Artificial Intelligence & Machine Learning
- Data Analytics
- System Design
- Problem Solving & Algorithms
- Database Management Systems
- Cloud Computing
- Product Engineering

TECHNICAL SKILLS

- **Programming & Problem Solving:** Java, Python, Data Structures & Algorithms, SQL
- **System Fundamentals:** DBMS, Operating Systems, Computer Networks
- **Full-Stack Development:** React.js, Spring Boot, REST APIs
- **AI & Machine Learning:** OpenCV, Pandas, Numpy, Prompt Engineering, YOLOv8
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Tools & Platforms:** Git, GitHub, Postman, Google Colab, VS Code, MySQL Workbench, Jupyter Notebook, Azure Data Studio, DB Browser, Canva, Docker (Basics)

CERTIFICATIONS

- SQL for Exploratory Data Analysis Essential Training
- HTML Essential Training – LinkedIn Learning
- Learning SQL Programming – LinkedIn Learning
- Participated — Pinnacle at CIT (Haptic Technologies)
- Pandas Bootcamp 2022: Complete Pandas Walkthrough - Udemy

SOFT SKILLS & HOBBIES

Soft Skills: Communication, Team Collaboration, Adaptability, Client Interaction, Analytical Thinking.

Hobbies: Listening Music, Playing Chess Online.